

Speech Recognition Spring 2025 Project 5

Due date: Feb 24th 2026 12pm Beijing time

In this assignment, you need to fill in the blanks at the specified locations and understand the core workflow, completing the training procedure of an end-to-end speech recognition system.

you are required to complete Stages 0 through 4 in 'run.sh'. You need to **report the loss and Word Error Rate (WER) metrics** on the validation dataset for the best checkpoint for the following tasks.

To monitor training progress, you can use TensorBoard to examine the learning curves for loss and WER. The logs for visualization are saved in the ./log directory.

You should explain how you fill in the blanks in the report.

TASK1

Train the model on librispeech dataset and use **CTC loss**. Set ctc_weight in the config file as 1.0.

TASK2

Train the model on librispeech dataset and use **attention loss**. Set ctc_weight in the config file as 0.0.

Tips:

1. This assignment requires a GPU to run. Please use dkucc to complete the task. The instruction file for using dkucc can be found in Project 2 materials.
2. The model may take a few hours to run, so running it directly in a terminal is not recommended, as accidentally closing the terminal could terminate the program. You can use "tmux" to start a persistent session for running your program. You can return to this session using "tmux a". More information on tmux usage can be found online.
3. This assignment references the complete project at <https://github.com/Alexander-H-Liu/End-to-end-ASR-Pytorch.git>. You may refer to the original code if you encounter any difficulties.